# American Computer Science League

2025-2026 ● Contest 3: Short Problems ● Elementary Division

**1. Boolean Algebra**

Which of the following Boolean expressions <u>cannot</u> be used to represent all 2-digit whole numbers?

A. x<100 AND x≥10
B. x≤99 AND x≥10
C. x<99 AND x>10
D. x≤99 AND x>9
E. x<100 AND x>9

**2. Boolean Algebra**

Which ordered pair makes the following expression TRUE?

$$\sim(A * B) * \sim(A + B)$$

A. (0,0)
B. (0,1)
C. (1,0)
D. (1,1)
E. None of the above

**3. Boolean Algebra**

Fill in the columns in the following Truth Table. Give the answer as the last column from top to bottom.

| $A$ | $B$ | $A + B$ | $\sim B$ | $\sim B * (A + B)$ | $\sim(\sim B * (A + B))$ |
|---|---|---|---|---|---|
| 0 | 0 | | | | |
| 0 | 1 | | | | |
| 1 | 0 | | | | |
| 1 | 1 | | | | |

A. 1000
B. 1101
C. 0001
D. 1001
E. 1010

**4. Boolean Algebra**

Simplify the following Boolean expression:

$$\sim(\sim A + B) + \sim(A + \sim B) + \sim(A * B)$$

A. $\sim(A * B)$
B. $A * \sim B + \sim A * B$
C. $\sim A * \sim B$
D. 1
E. 0

| | |
|---|---|
| **5. Boolean Algebra**<br><br>The Boolean algebra operator $A \rightarrow B$ is often read as "if A, then B" or "A implies B". It is TRUE for the ordered pairs (1,1), (0,1), and (0,0). Which of the following is an equivalent expression to $A \rightarrow B$? | A. $\sim(A + B)$<br>B. $\sim A * \sim B$<br>C. $A + \sim B$<br>D. $\sim A + B$<br>E. $\sim A * \sim B + A * B$ |
| **6. Boolean Algebra**<br><br>In a local elementary school of 100 students, 50 students play an instrument and 60 students play a sport while 20 students play both. If I = {students who play an instrument} and S = {students who play a sport}, how many students would the expression $\sim(I + S)$ represent? | A. 90<br>B. 80<br>C. 50<br>D. 20<br>E. 10 |

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**1. 布尔代数**

**请问**以下哪个布尔表达式<u>不能</u>用来表示所有的两位整数？

A. x<100 AND x≥10
B. x≤99 AND x≥10
C. x<99 AND x>10
D. x≤99 AND x>9
E. x<100 AND x>9

**2. 布尔代数**

**请问哪个有序对可以使得下列表达式的结果为真？**

$$\sim(A * B) * \sim(A + B)$$

A. (0,0)
B. (0,1)
C. (1,0)
D. (1,1)
E. **上述答案均不正确**

**3. 布尔代数**

**填写以下真值表，将最后一列的内容按照从上到下的顺序依次写出结果。**

| $A$ | $B$ | $A + B$ | $\sim B$ | $\sim B * (A + B)$ | $\sim(\sim B * (A + B))$ |
|---|---|---|---|---|---|
| 0 | 0 | | | | |
| 0 | 1 | | | | |
| 1 | 0 | | | | |
| 1 | 1 | | | | |

A. 1000
B. 1101
C. 0001
D. 1001
E. 1010

**4. 布尔代数**

请简化**下列布尔表达式：**

$$\sim(\sim A + B) + \sim(A + \sim B) + \sim(A * B)$$

A. $\sim(A * B)$
B. $A * \sim B + \sim A * B$
C. $\sim A * \sim B$
D. 1
E. 0

**5. 布尔代数**

布尔**代数运算符** $A \rightarrow B$ **通常读作“若** A**，则** B**”或“**A **蕴含** B**”。对于有序数对** (1,1), (0,1), 和 (0,0)**，该运算结果为真。下列哪一项表达式与** $A \rightarrow B$ **等价？**

A. $\sim(A + B)$
B. $\sim A * \sim B$
C. $A + \sim B$
D. $\sim A + B$
E. $\sim A * \sim B + A * B$

**6. 布尔代数**

某当地小学有 100 名学生，其中 50 名学生演奏乐器，60 名学生参加体育运动，20 名学生两项都参与。设 I = {演奏乐器的学生集合}，S = {参加体育运动的学生集合}。**请问**表达式 $\sim(I + S)$ 表示多少名学生？

A. 90
B. 80
C. 50
D. 20
E. 10